

Do Politicians' Words have a Dollar Value?

*Measuring the Impact of Climate Policy Speeches on
Green Bond Prices in India, Indonesia, and Thailand*

Research Proposal

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1. Introduction: The Greenium in Emerging Economies

In the world of finance, “*Green Bonds*” are a special type of loan used by governments to fund environmental projects. When investors trust a government’s climate commitment, they often accept a lower interest rate on these bonds—a discount known as the “**Greenium**”. It is defined as the yield spread between a sovereign green bond and an otherwise identical conventional bond:

$$\text{Greenium} = y_{\text{green}} - y_{\text{conventional}} \quad (1)$$

where a negative value represents the cost-saving “premium” for the government. For emerging economies like **India, Indonesia, and Thailand**, this discount is not just a technicality; it is a vital financial tool that saves millions of dollars in debt payments, providing the “fiscal breathing room” needed to transition away from fossil fuels.

While the Greenium is well-studied in Europe and the US, it remains highly volatile in Asian emerging markets. This research focuses on the **secondary market** (daily trading) in India, Indonesia, and Thailand from 2020 to 2025. These countries are “Transition Giants”—they have massive coal dependencies but are also leading the world in new Green Bond issuances. The scope is specifically narrowed to how **daily political communication** (speeches and news) moves these bond prices, rather than just looking at slow-moving economic data like GDP.

The primary contribution of this thesis is to prove that **political consistency is a financial asset**. By using AI to turn thousands of news headlines into a “Sentiment Score,” I will demonstrate that investors act as a “truth-detector”:

- If the research shows a strong link—termed a “*Credibility Premium*”—it proves that “Green Talk” directly lowers a nation’s borrowing costs.
- If the link is weak, it proves that a “*Credibility Gap*” exists, where investors ignore political promises due to a lack of trust.

Either way, the study provides a first-of-its-kind “Policy Recommendation”: to lower national debt, a government must synchronize its climate message to ensure it is perceived as credible by global markets. Specifically, this study addresses three objectives: (i) the high-frequency measurement of sentiment, (ii) the causal impact of political shocks, and (iii) the market’s reaction to signaling noise.

Research Questions

1. *Are global investors actually listening to what Asian politicians say about climate change?*
In simpler terms: Can we accurately turn news into a “Sentiment Score” using AI?
2. *When a politician makes a green promise, does the government actually save money on its loans?*
In simpler terms: Do “shocks” (big speeches) cause the Greenium to widen or shrink?
3. *Does it pay to be honest and consistent?*
In simpler terms: How does the market penalize “signaling noise”?

2. Novelty & Literature Review

2.1. The “Greenium” Debate: Consensus vs. Skepticism

The existence of a “Greenium”—the yield discount investors accept for green bonds—is a foundational but highly contested topic in modern finance. Early research by [Zerbib \(2019\)](#) and [Baker et al. \(2022\)](#) established that “pro-environmental preferences” allow green issuers to borrow at lower rates than conventional counterparts. In the specific context of “Transition Giants,” [Panizza et al. \(2025\)](#) recently identified a significant 13 basis point Greenium in emerging markets, suggesting these bonds act as a hedge against the economic dangers of shifting away from fossil fuels.

However, a growing body of literature challenges this consensus. [Larcker and Watts \(2020\)](#) found a “Zero Greenium” in municipal markets, arguing that if risk profiles are identical, rational investors view green and conventional bonds as “exact substitutes.” Furthermore, [Flammer \(2021\)](#) suggests that a green label alone is insufficient; without a strong sovereign reputation, the market simply ignores the signal. [Neumann \(2025\)](#) builds on this by arguing that in countries like India, Indonesia, and Thailand, green progress is often “stalled” by domestic coal interests, creating a “Credibility Gap” that standard monthly indices—such as the Economic Policy Uncertainty (EPU) index—fail to capture.

2.2. Novelty: Resolving the Debate through Signaling

The novelty of this study lies in using Natural Language Processing (NLP) to act as a “tie-breaker” between these conflicting theories. While traditional IPE research treats political risk as static background noise, this study treats it as a financial signal.

1. **Methodological Granularity:** By utilizing the **GDELT 2.0** database and the **ClimateBERT** model, I move beyond “blunt” monthly metrics to measure daily “Political Shocks.” This aligns with recent findings by [Dragotto et al. \(2025\)](#), who demonstrated that high-frequency climate awareness is a primary driver of market dynamics.

2. **Quantifying Policy Credibility:** I bridge the gap between “Green Talk” and market value. Following the framework of [Lau et al. \(2024\)](#), I test whether “Policy Credibility” is the missing variable that determines if a bond receives a discount or faces a “Trust Penalty.”
3. **Testing “Signaling Noise”:** I introduce a unique test for consistency—measuring the market’s reaction when a government’s climate promises conflict with its industrial actions.
4. **Geographic Focus:** Focusing on India, Indonesia, and Thailand provides the first causal evidence of how localized political rhetoric impacts borrowing costs in the world’s most volatile green markets.

3. Data Collection Method

To calculate the Greenium and link it to political sentiment, we require two primary datasets: **Daily Bond Yields** and **Political News**.

3.1. Financial Data: Daily Bond Yields (YTM)

We need the daily *Yield to Maturity* (YTM) for both sovereign green bonds and their matched conventional “twin” bonds.

- **Source:** AsianBondsOnline (ADB) and Investing.com.
- **Collection:** AsianBondsOnline is a free data portal managed by the Asian Development Bank. It provides historical CSV downloads for local currency (LCY) government bonds.
- **The “Twin” Selection:** For each country, we will select:
 1. *The Green Bond:* (e.g., India’s 2023 Sovereign Green Bond).
 2. *The Conventional Bond:* A “normal” bond with a similar maturity date (e.g., if the green bond expires in 2033, we pick a normal bond that also expires in 2033).
- **Cost:** Free.

3.2. Sentiment Data: Global Political News

We need to capture every major political news story or speech related to climate change in our target countries.

- **Source:** The GDELT Project (Global Database of Events, Language, and Tone).
- **Collection:** GDELT is a free, Google-supported database that monitors the world’s news in 100+ languages. We will use the **GDELT 2.0 Doc API** (via a Python script) to filter for:
 - *Geography:* SourceCountry = “IN” (India), “ID” (Indonesia), “TH” (Thailand).

- *Themes*: “CLIMATE_CHANGE”, “RENEWABLE_ENERGY”, “ENVIRONMENT”.

- **Cost**: Free.

3.3. Processing: Sentiment Scoring via ClimateBERT

Raw news headlines are just text; we need to convert them into a “Sentiment Score” ranging from -1 to $+1$.

- **Tool**: ClimateBERT (via Hugging Face).
- **How it works**: Daily GDELT headlines are fed into ClimateBERT. The AI labels each day’s news sentiment:
 - *Positive* ($+1$): “Prime Minister announces \$10B solar investment.”
 - *Negative* (-1): “Government approves 5 new coal-fired power plants.”
- **Cost**: Free (open-source model).

4. Methodology

To isolate the impact of political signaling from general market noise, the research follows a two-stage econometric approach.

4.1. The Matching Method (Creating the Greenium)

To ensure that yield differences are not driven by maturity risks or coupon effects, I follow the “Synthetic Twin” approach established by [Zerbib \(2019\)](#).

1. **Selection**: For each sovereign green bond in India, Indonesia, and Thailand, I identify two conventional (non-green) bonds from the same issuer with the closest maturity dates.
2. **Linear Interpolation**: A “Synthetic Conventional Yield” is calculated by interpolating between the two conventional bonds to match the exact maturity of the green bond.
3. **Calculation**: The daily Greenium is then:

$$\text{Greenium}_{c,t} = y_{\text{green},c,t} - y_{\text{conventional},c,t} \quad (2)$$

4.2. NLP Sentiment Quantification

I convert qualitative political “noise” into quantitative financial data using a specialized Natural Language Processing (NLP) pipeline.

1. **Extraction**: Using the **GDELT 2.0 API**, I scrape all daily news headlines and official government statements containing climate-related keywords (e.g., “Paris Agreement,”

“Net Zero,” “Coal Phase-out”) for the three target countries. I identify “shocks” as days with high-intensity signaling. Examples include:

- *Regulatory Shocks*: The release of India’s Sovereign Green Bond Framework or Thailand’s Green Taxonomy.
- *International Pledges*: Indonesia’s launch of the Just Energy Transition Partnership (JETP) at the G20.
- *Contradictory Signals (Signaling Noise)*: Government approvals for new coal-fired plants in East Kalimantan or coal-railway expansions in Chhattisgarh, which conflict with “Net Zero” rhetoric.

2. **Classification**: I employ **ClimateBERT**, a transformer model pre-trained on environmental text, to assign a daily “Climate Sentiment Score” ($\text{Sentiment}_{c,t}$). This score ranges from -1 (contradictory/brown signals) to $+1$ (credible/green signals).

4.3. Difference-in-Differences (DiD) and Event Study Framework

In this framework, the “treatment” is defined as discrete **Political Sentiment Shocks** (major speeches or policy shifts) that occur at different times across the three countries.

- **Treatment Group**: Sovereign Green Bonds ($\text{Green}_i = 1$), hypothesized to be hypersensitive to climate-related political signaling.
- **Control Group**: The **Synthetic Conventional Twins** ($\text{Green}_i = 0$), serving as the counterfactual by representing how the market reacts to the same country-level news in the absence of a green mandate.
- **Event Study Logic**: Following the high-frequency event study logic of [Dragotto et al. \(2025\)](#), I utilize a **24-hour “Pre-Treatment” window** to establish the baseline and a **24-hour “Post-Treatment” window** to capture the immediate price discovery. This allows me to test the **Parallel Trend Assumption**: that in the absence of the speech, the Green Bond and its Synthetic Twin would have continued to follow a similar trend.
- **Robustness of the Window**: Recognizing that emerging markets may exhibit delayed reactions due to time-zone differences or lower liquidity, I will also test **48-hour and 72-hour windows** as robustness checks to ensure the results are not sensitive to the specific timeframe chosen.

The Fixed Effects Panel Regression Model is specified as:

$$\Delta y_{it} = \alpha_i + \delta_t + \beta_1 (\text{Green}_i \times \text{Post}_t) + \beta_2 \text{Sentiment}_{c,t} + \gamma X_{ct} + \varepsilon_{it} \quad (3)$$

where:

- α_i (**Bond Fixed Effects**): Controls for all time-invariant characteristics of the specific bond (e.g., currency, country risk, original contract terms).
- δ_t (**Time Fixed Effects**): Filters out “Global Noise” by absorbing daily global shocks such as US Federal Reserve interest rate hikes or shifts in global oil prices, ensuring the model only captures local political signals.
- β_1 (**Interaction Term**): Measures the causal “jump” in the Greenium following specific major political announcements (e.g., the launch of a National Green Taxonomy).
- β_2 (**Sentiment Coefficient**): Estimates the marginal effect of daily “Green Talk” on the cost of sovereign debt.

4.4. Testing the “Credibility Gap” (Robustness)

For robustness, I will perform a sub-sample analysis comparing “*Harmonized Signaling*” periods (where rhetoric matches policy action) against “*Signaling Noise*” periods (conflicting messages). If the Greenium only widens during harmonized periods, it provides empirical proof of a “**Trust Penalty**” in the market.

Limitations

1. **Investor Heterogeneity:** While this study focuses on the aggregate secondary market reaction, it is acknowledged that domestic and foreign investors may respond differently to local political signals due to varying levels of “home bias” and familiarity with local political nuances. However, distinguishing between these investor classes requires granular custodial data (e.g., EPFR data), which is outside the current scope of this paper. This remains a significant avenue for future research.

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